



Comparison of Two Models with Non-ordinary Arrival Process to Analyze the Cluster of the DECT-2020 Network

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Abstract. DECT-2020 NR is a wireless communication standard designed for large-scale IoT networks with a large number of devices. In such networks, a sink node collects data from multiple sources. In this paper, we modeled a DECT-2020 sink node as a discrete packet arrival system, where data packets are processed in batches. We compared two types of systems with different arrival disciplines, Early Arrival and Late Arrival, across several metrics, including the age of information (AoI), which measures the freshness of the received data. The results showed how the difference in the order of batch arrival affected network performance and Age of Information.

Keywords: DECT-2020 · URLLC · mMTC · age of information · queuing theory · discrete time Markov chain

1 Introduction

The rapid development of fifth-generation (5G) networks creates new opportunities for enabling applications that demand massive machine-type communications (mMTC) and ultra-reliable low-latency communications (URLLC). These capabilities are of particular importance in the context of the Internet of Things (IoT), where millions of devices are expected to communicate simultaneously with high levels of reliability, resilience, and scalability.

Among the promising technologies in this domain is the DECT-2020 New Radio (NR) protocol, standardized by the European Telecommunications Standards Institute (ETSI) [1–4]. Unlike conventional cellular systems that rely on centralized base stations and fixed infrastructure, DECT-2020 NR operates in the unlicensed 1880–1930 MHz frequency band and supports a fully decentralized mesh network topology (see Fig. 1).

In DECT-2020 NR, each radio device (RD) can act as a transmitter (RD_{PT}), repeater ($RD_{FT,PT}$), or receiver (RD_{FT}), thereby supporting flexible multi-hop communication. This self-organizing architecture enables scalable deployments with densities of up to one million devices per square kilometer [5], which significantly exceeds the capacity of traditional wireless technologies such as Wi-Fi, LTE-M, or NB-IoT.

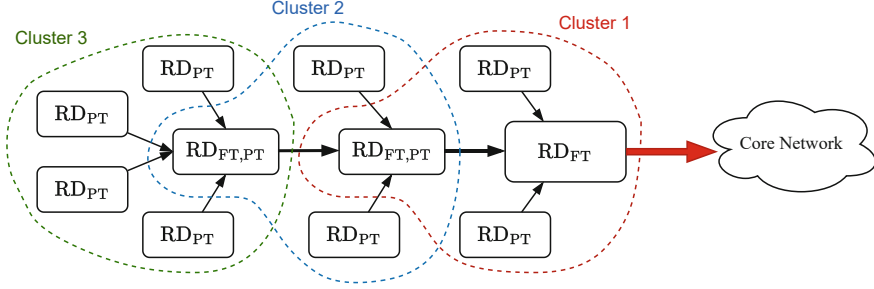


Fig. 1. DECT-2020 network topology

Another important characteristic of DECT-2020 NR is its half-duplex transmission mode. In this case, a relay node ($RD_{FT,PT}$) alternates between receiving packets from child nodes (RD_{PT} or $RD_{FT,PT}$) and transmitting packets to the parent node ($RD_{FT,PT}$ or RD_{FT}). Such an operation can be naturally represented by a queueing model with batch arrivals and batch services, where the arrival batch size is variable while the service batch size remains fixed.

When modelling a tact-based systems such as DECT-2020 NR, it is important to have a clear understanding of the sequence of actions in a slot, which is especially important in half-duplex data transmission mode. Specialized literature distinguishes between two main disciplines: Early Arrival (EA) and Late Arrival (LA) [6]. As the names suggest, in the first case, a data packet is accepted at the beginning of the slot and transmitted further through the system at the end of the slot. In the second case, the data packet stored on the node is first transmitted through the system, and only at the end of the slot the new data packet arrives. In addition to DECT-2020 NR, EA and LA can also be used in IAB technology, which can also operate in half-duplex data transmission mode.

This paper focuses on the modeling and performance evaluation of a DECT-2020 NR network cluster using discrete-time queueing systems. Special attention is given to the analysis of packet transmission delays in multi-hop routing scenarios and the influence of the stochastic nature of mMTC traffic on overall network performance.

This article proposes two mathematical models of the DECT-2020 NR cluster. To this end, Sect. 2 constructs a mathematical model of packet transmission by a relay node ($RD_{FT,PT}$) in the form of two different queueing systems, taking into account the peculiarities of mMTC traffic, i.e. multiple random access.

Section 3 describes a numerical experiment illustrating the difference between two mathematical models applied to a real DECT-2020 NR node.

Related Works. In recent years, discrete-time queues have been attracting much attention, partly due to their practical relevance. Dividing time into slots and the need to select different arrival disciplines (EA or LA) allows for more accurate modeling of telecommunication networks.

Earlier, within the framework of DTMC studies, authors compared and analyzed models with different arrival policies. In the article by Gupta and Goswami [6], a discrete-time system with a finite buffer and bulk service was analyzed. The authors considered two policies—AF (arrival first) and DF (departure first)—showing that AF reduces delays and losses, while DF can be effective under high load. Similarly, the paper [7] examined the relationships between queue lengths at different epochs (arrival, departure, and random) for a discrete-time queue with D-BMAP arrivals, also under different arrival policies (EA and LA). The authors used properties of the discrete-time batch Markovian arrival process (D-BMAP) and probability generating function methods to establish exact relationships between distributions. The results allow characteristics at one epoch to be expressed in terms of another, simplifying system analysis. The article [8] by Goswami and Chaudhry presents the first explicit analysis of the sojourn time distribution for a discrete-time queue with batch service $Geo/Geo^{a,b}/1$. The authors propose a novel analytical approach, utilizing the roots of the characteristic equation inside and outside the unit circle, to derive closed-form expressions for the sojourn time probability generating function and its probability distribution. Research has also been conducted on systems with bulk arrivals and bulk service. The article by Claeys [9] focused on delay analysis in the discrete-time $Geo^X/Geo^c/1$ system with bulk arrivals and bulk service. The authors considered two service policies: IBSP (immediate-batch service policy) and FBSP (full-batch service policy). Additionally, Chang and Choi [10] analyzed a discrete-time system with a finite buffer where the server operates in bulk arrival and bulk service mode and can also take vacations. Discrete-time systems are also actively used in the study of Age of Information (AoI). For example, in the study [11] the authors obtained the stationary distribution of AoI and derived analytical expressions for the average age of information and other performance metrics using a discrete-time queueing system.

In contrast to the works presented above, the authors of this article analyze a discrete-time system with batch arrivals of variable size and batch service of fixed size under EA and LA disciplines. For both systems, explicit formulas for the probability generating function of the waiting time until the start of service were derived.

2 System Model

In this section, for the system model of the node $\text{RD}_{\text{FT,PT}}$ we built the mathematical model in the form of a discrete-time queueing system. We complete this section by introducing the metrics of interest. The notation utilized in this paper is provided in Table 1.

Table 1. The notation used in the paper

Notation	Description
A	maximum length of an arrival batch
B	constant length of a service batch
$l_i, i = \{1, \dots, A\}$	probability distribution of the arrival batch size
$\hat{l}$	the average number of packets in an arrival batch
L_n	the probability that an arrival batch has at least n packets
$L(z)$	PGF of the arrival batch size
a	probability of a batch arrival
b	probability of a completion of a batch service
ρ	system load
$Q^-(n)$	discrete time Markov chain (DTMC) of the number of packets in the system
Ω	discrete state space of DTMC $Q^-(n)$
p_{ij}^{la}, p_{ij}^{ea}	transition probabilities of $Q^-(n)$
p_i^{la}, p_i^{ea}	stationary probabilities of $Q^-(n)$
$P_{LA}(z), P_{EA}(z)$	PGFs of stationary probabilities of $Q^-(n)$
$Q^*(n)$	the embedded DTMC
q_i^{la}, q_i^{ea}	stationary probabilities of $Q^*(n)$
$Q_{LA}(z), Q_{EA}(z)$	PGFs of stationary probabilities of $Q^*(n)$
$\alpha(z)$	PGF of the batch inter-arrival time
$\beta(z)$	PGF of the batch service time
$\mathcal{W}_{LA}(z), \mathcal{W}_{EA}(z)$	PGFs of a packet waiting time
$V_{LA}(z), V_{EA}(z)$	PGFs of a packet sojourn time
$\bar{w}_{LA}, \bar{w}_{EA}$	the average waiting time of a packet
$\bar{v}_{LA}, \bar{v}_{EA}$	the average sojourn time of a packet
N_{LA}, N_{EA}	the average number of packets in the system
$\Delta(t)$	function of Age of Information
Δ_i^P	PAoI for the i th packet
$\Delta^P(z), \Delta_{LA}^P(z), \Delta_{EA}^P(z)$	PGFs of PAoI
$\bar{\Delta}^P, \bar{\Delta}_{LA}^P, \bar{\Delta}_{EA}^P$	the average PAoI

The DECT-2020 network cluster consists of D terminal devices RD_{PT} with one relay $\text{RD}_{\text{FT,PT}}$ as repeater node or the sink RD_{FT} as the final receiver node, in which the age of information from the terminal device RD_{PT} transmitted in the uplink towards the base network is estimated. The system model of the DECT-2020 network considered in this work is based on the previously published study [12]. At specified by half duplex mode time intervals, each terminal device generates a data packet and attempts to transmit it to the receiver via repeater, where the packets await further transmission along the route. The number of packets received by $\text{RD}_{\text{PT,FT}}$ node is a random variable (r.v.), the distribution of which depends on the number D of terminal devices RD_{PT} . The repeater $\text{RD}_{\text{FT,PT}}$ also has specified time intervals to transmit packets up to the parent node. The number of packets transmitted by node $\text{RD}_{\text{PT,FT}}$ during the transmission time interval is also random, but in the most congested links, such as the link from RD_{FT} to the core network, it can be considered constant, and its value is determined by the duration of the transmission interval.

2.1 Queueing System

We present the mathematical model of the $\text{RD}_{\text{FT,PT}}$ node as an infinite buffer discrete-time queue with batch arrivals of variable size and batch service of fixed size, formally denoted as $\text{Geo}^{[1,A]}|\text{Geo}^{[B]}|1|\infty$. Packets arrive in batches of no more than A packets, that is given by D , and are served in batch with fixed size B , which corresponds to the average number of packets transmitted per time slot, depended on the channel data rate.

Batches of packets arrive in the system with the probability a per slot and no packets arrive with the complementary probability $\bar{a} = 1 - a$. Probability a represents a probability of generating a data packet by at least one RD_{PT} from D RD_{PT} . Probability b means that the batch service is completed during a slot, and the probability that no batch completes the service during a slot, equals to $\bar{b} = 1 - b$. Probability b allows us to model the successful completion of a packet batch by node RD_{FT} at the end of the transmission interval. These allow modeling of the half-duplex communication mode.

In this paper, we consider two types of discrete-time system: Late Arrival (LA) and Early Arrival (EA) [6]. There is difference in the order of actions in a slot, see Fig. 2:

- *LA system*: at the beginning of the slot a batch of packets completes service and leaves the system, then, depending on the number of packets remaining in the queue, the next batch of size B either begins service immediately or in one of the next time slots once B packets have accumulated in the queue and at the end of the slot new batch of packets arrives;
- *EA system*: at the beginning of the slot new batch of packets arrives, then if a server is free depending on the number of packets remaining in the queue immediately or in the next time slot the batch of B packets begins service and at the end of the slot leaves the system.

Next, we will introduce some necessary definition which are the same for the both systems. After that, we will derive formulas for each systems, separately.

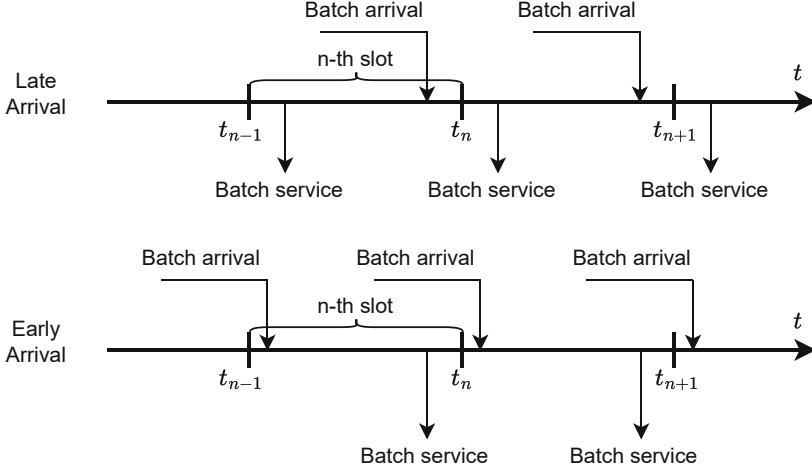


Fig. 2. The time diagram for two considered system: Late Arrival and Early Arrival

The following notations for both LA and EA are introduced:

$\{l_i, i = 1, \dots, A\}$ - probability distribution of the arrival batch size;

$L(z) = \sum_{i=1}^A l_i z^i$ - probability generating function (PGF) of the arrival batch size;

$\hat{l} = \sum_{i=1}^A i l_i$ - the average number of packets in an arrival batch.

Let us define the discrete time Markov chain (DTMC) $Q^-(n), n = 0, 1, \dots$, with discrete state space $\Omega \in \{0, 1, \dots\}$ denoting the number of packets in the system immediately before the end of the n th slot after the batch arrived.

Also, we note that steady-state conditions for both LA and EA systems exist only if the following is satisfied

$$\rho = \frac{a\hat{l}}{bB} < 1. \quad (1)$$

Early Arrival System. We define the stationary probability distribution of the system states as

$$p_i^{ea} = \lim_{n \rightarrow \infty} P\{Q^-(n) = i\}, \quad i \in \Omega, \quad (2)$$

and to find them we determine the transition probabilities of DTMC $Q^-(n)$

$$p_{ij}^{ea} = P\{Q^-(n) = j | Q^-(n-1) = i\} =$$

$$= \begin{cases} \bar{a} + al_B b, & i = j < B, & (3a) \\ al_{j-i} + al_{j-i+B} b, & i < j \leq B-1, j-i \leq A-B & (3b) \\ al_{j-i}, & i < j \leq B-1, j-i > A-B & (3c) \\ \bar{a}\bar{b} + al_B b, & j = i \geq B, & (3d) \\ \bar{a}b, & i - j = B, & (3e) \\ al_{j-i+B} b, & i > j, i - j < B, & (3f) \\ al_{j-i}\bar{b} + al_{j-i+B} b, & j - i \leq A-B, j > i, & (3g) \\ al_{j-i}\bar{b}, & j - i > A-B, j > i, & (3h) \\ 0, & \text{otherwise.} \end{cases}$$

Expression (3a) represents the probability that the number of packets in the system remains unchanged if initially there were fewer than B packets in the system.

(3b)—the number of packets increases by $j - i$; either exactly $j - i$ packets arrive and none are serviced, or more than $j - i$ packets arrive and the B packets already in the system are serviced.

(3c)—the number of packets increases by $j - i$, meaning exactly $j - i$ packets arrive and none are serviced.

(3d)—the number of packets remains unchanged if initially there were more than B packets or exactly B packets in the system.

(3e)—no packets arrive and exactly B packets are serviced.

(3f)—the number of packets increases by $j - i$, meaning $j - i + B$ packets arrive and B packets are serviced.

(3g)—the number of packets increases by $j - i$; either $j - i + B$ packets arrive and B packets are serviced, or only $j - i$ packets arrive and none are serviced.

(3h)—the number of packets increases by $j - i$ and none are serviced.

$$\begin{aligned} p_i^{ea} &= \sum_{j=0}^{i-A+B-1} p_j^{ea} al_{i-j} + \sum_{j=\max(0, i-A+B)}^{i-1} p_j^{ea} (al_{i-j} + al_{i-j+B} b) + \\ &+ p_i^{ea} (\bar{a} + al_B b) + \sum_{j=i+1}^{B+i-1} p_j^{ea} al_{i-j+B} b + p_{i+B}^{ea} \bar{a}b, \quad 0 \leq i < B, \\ p_i^{ea} &= \sum_{j=\max(0, i-A)}^{i-A+B-1} p_j^{ea} al_{i-j} \bar{b} + \sum_{j=i-A+B}^{i-1} p_j^{ea} (al_{i-j} \bar{b} + al_{i-j+B} b) + \\ &+ p_i^{ea} (\bar{a}\bar{b} + al_B b) + \sum_{j=i+1}^{B+i-1} p_j^{ea} al_{i-j+B} b + p_{i+B}^{ea} \bar{a}b, \quad i \geq B. \end{aligned} \quad (4)$$

The system of Eq. (4) can be solved numerically, but for the case of an infinite buffer, only an approximate numerical solution is achievable. However, in this case one can obtain an explicit analytical expression for the generating function

$P_{EA}(z)$ of the stationary probability distribution (2) for the EA system:

$$P_{EA}(z) = \sum_{i=0}^{\infty} p_i^{ea} z^i. \quad (5)$$

By multiplying (4) by z^i and then summing up over i , we have

$$P_{EA}(z) = \frac{b(z^B - 1) \sum_{i=0}^{B-1} p_i^{ea} z^i \left(\bar{a} + a \sum_{k=1}^{B-i-1} z^k l_k \right)}{z^B - (\bar{a} + aL(z))(\bar{b}z^B + b)}. \quad (6)$$

Refer to the Appendix for the omitted details of derivation.

According to Rouché's theorem [16] and using the normalization condition, we can acquire $p_i^{ea}, i = 0, \overline{B-1}$, due to the next system of equations:

$$\begin{cases} \sum_{i=0}^{B-1} p_i^{ea} z_j^i \left(\bar{a} + a \sum_{k=1}^{B-i-1} z_j^k l_k \right) = 0, & j = \overline{1, B-1}, \\ \sum_{i=0}^{B-1} p_i^{ea} \left(\bar{a} + a \sum_{k=1}^{B-i-1} l_k \right) = 1 - \rho, \end{cases} \quad (7)$$

where $z_j, j = \overline{1, B-1}$, are the roots of the denominator of (6), which is $|z_j| < 1$.

Late Arrival System. As for the LA system, we define the stationary probability distribution

$$p_i^{la} = \lim_{n \rightarrow \infty} P\{Q^-(n) = i\}, \quad i \in \Omega, \quad (8)$$

and determine the transition probabilities of DTMC $Q^-(n)$

$$p_{ij}^{la} = P\{Q^-(n) = j | Q^-(n-1) = i\} =$$

$$= \begin{cases} \bar{a}, & i = j < B, & (9a) \\ al_{j-i}, & i < j, i = \overline{0, B-1}, j = \overline{1, A} & (9b) \\ \bar{a}\bar{b} + bal_B, & j = i \geq B, & (9c) \\ b\bar{a}, & i - j = B, & (9d) \\ bal_{j-i+B}, & i > j, i - j < B, & (9e) \\ \bar{b}al_{j-i} + bal_{j-i+B}, & j - i \leq A - B, j > i, & (9f) \\ \bar{b}al_{j-i}, & j - i > A - B, j > i, & (9g) \\ 0, & \text{otherwise.} \end{cases}$$

Similarly to (3), in (9) expression (9a) represents the probability that the number of packets in the system remains unchanged if initially there were fewer than B packets in the system.

(9b)—the number of packets increases by $j - i$, meaning exactly $j - i$ packets arrive and none are serviced.

(9c)—the number of packets remains unchanged if initially there were more than B packets or exactly B packets in the system.

(9d)—no packets arrive and exactly B packets are serviced.

(9e)—the number of packets increases by $j - i$, meaning $j - i + B$ packets arrive and B packets are serviced.

(9f)—the number of packets increases by $j - i$; either $j - i + B$ packets arrive and B packets are serviced, or only $j - i$ packets arrive and none are serviced.

(9g)—the number of packets increases by $j - i$ and none are serviced.

The stationary probability distribution can be obtained as shown in (10), using the transition probabilities (9), but LA system has the same situation with the problem to solve this system of equation directly as EA system, so here we also utilize the method of generating functions.

$$\begin{aligned}
 p_i^{la} &= \sum_{j=0}^{i-1} p_j^{la} a l_{i-j} + p_i^{la} \bar{a} + \sum_{j=B}^{B+i-1} p_j^{la} b a l_{i-j+B} + p_{i+B}^{la} b \bar{a}, \quad 0 \leq i \leq B-1, \\
 p_i^{la} &= \sum_{j=0}^{B-1} p_j^{la} a l_{i-j} + \sum_{j=B}^{i-1} p_j^{la} (\bar{b} a l_{i-j} + b a l_{i-j+B}) + p_i^{la} (\bar{b} \bar{a} + b a l_B) + \\
 &\quad + \sum_{j=i+1}^{B+i-1} p_j^{la} b a l_{i-j+B} + p_{i+B}^{la} b \bar{a}, \quad i = \overline{B, A}, \\
 p_i^{la} &= \sum_{j=i-A}^{B-1} p_j^{la} a l_{i-j} + \sum_{j=\max(B, i-A)}^{i-A+B-1} p_j^{la} \bar{b} a l_{i-j} + \sum_{j=i-A+B}^{i-1} p_j^{la} (\bar{b} a l_{i-j} + b a l_{i-j+B}) + \\
 &\quad + p_i^{la} (\bar{b} \bar{a} + b a l_B) + \sum_{j=i+1}^{B+i-1} p_j^{la} b a l_{i-j+B} + p_{i+B}^{la} b \bar{a}, \quad i > A.
 \end{aligned} \tag{10}$$

We define the PGF of the stationary probabilities in (8) as

$$P_{LA}(z) = \sum_{i=0}^{\infty} p_i^{la} z^i. \tag{11}$$

By multiplying (10) by z^i and then summing up over i , we have

$$P_{LA}(z) = \frac{b(\bar{a} + aL(z))(z^B - 1) \sum_{i=0}^{B-1} p_i^{la} z^i}{z^B - (\bar{a} + aL(z))(\bar{b}z^B + b)}. \tag{12}$$

Refer to the Appendix for the omitted details of derivation.

According to Rouché's theorem [16], the denominator of (12) has $A+B$ roots $z_j, j = \overline{0, A+B}$ that include $z_0 = 1, |z_j| < 1, j = \overline{1, B-1}$, and $|z_j| > 1, j = \overline{B, A+B}$. We can also obtain an equation using the normalization condition:

$$\sum_{i=0}^{B-1} p_i^{la} = 1 - \rho. \quad (13)$$

With that, we form the system of linear equations by substituting the corresponding roots $z_j, |z_j| < 1, j = \overline{1, B-1}$, into the numerator (12), equating to zero:

$$\begin{cases} \sum_{i=0}^{B-1} p_i^{la} z_j^i = 0, & j = \overline{1, B-1}, \\ \sum_{i=0}^{B-1} p_i^{la} = 1 - \rho, \end{cases} \quad (14)$$

and solving this, we get

$$\sum_{i=0}^{B-1} p_i^{la} z^i = (1 - \rho) \frac{(-1)^{B-1} \prod_{i=1}^{B-1} (z - z_i)}{\prod_{i=1}^{B-1} (z_i - 1)}. \quad (15)$$

Finally, we acquire $P_{LA}(z)$ as

$$P_{LA}(z) = \frac{(-1)^{B-2} b(\bar{a} + aL(z))(1 - \rho) \sum_{i=0}^{B-1} z^i}{a\bar{b}l_A \prod_{j=B}^{A+B} (z - z_j) \prod_{j=1}^{B-1} (z_j - 1)}, \quad (16)$$

where $z_j, j = \overline{1, B-1}$, are the roots of the denominator of (12), which is $|z_j| < 1$, and $z_j, j = \overline{B, A+B}$, are such that $|z_j| > 1$.

It can be noticed that the same roots are used for both systems because of the same denominator in PGFs (6) and (12). The difference is only in the numerator of PGFs and in the incomplete form of the $L(z)$, in particular.

2.2 Metrics of Interest

We now proceed to derive the formulas for the PGF of sojourn time for a packet in the system for the LA system, then, by analogy, we will find one for the EA system.

First, we define the second DTMC $Q^*(n)$, $n = 0, 1, \dots$, with the same discrete state space Ω , whose states represent the number of packets in the queueing system immediately before the arrival of a new batch of packets at the n th time

slot. Let $\{q_i^{la}, i \in \Omega\}$ denote the stationary probability distribution of the states of the DTMC $Q^*(n)$, $n = 0, 1, \dots$:

$$q_i^{la} = p_i^{la} + p_{i+B}^{la}b, \quad i = \overline{0, B-1}, \quad (17)$$

$$q_i^{la} = p_i^{la}\bar{b} + p_{i+B}^{la}b, \quad i \geq B. \quad (18)$$

We also use the PGF method to find (17) and get the PGF of the corresponding distribution $Q_{LA}(z) = \sum_{i=0}^{\infty} q_i^{la} z^i$:

$$Q_{LA}(z) = P_{LA}(z) \left(\bar{b} + \frac{b}{zb} \right). \quad (19)$$

Before proceeding to the calculation of the PGF of the waiting time, we introduce the PGF of inter-arrival time and the PGF of the service time, respectively, $\alpha(z) = az/(1 - \bar{a}z)$, $\beta(z) = bz/(1 - \bar{b}z)$. These PGFs will be used for EA system too.

Now we are able to define the PGF of waiting time until the start of service for an arbitrary packet in an arriving batch

$$\mathcal{W}_{LA}(z) = \begin{cases} \sum_{i=0}^{\infty} q_i^{la} \sum_{k=1}^A \frac{l_k}{\bar{l}} \left[\sum_{j=1}^k \beta^{\gamma_i}(z) \alpha^{L_{\psi_i+k}}(z) \right], & k < \psi_i, \\ \sum_{i=0}^{\infty} q_i^{la} \sum_{k=1}^A \frac{l_k}{\bar{l}} \left[\sum_{j=1}^{\psi_i} \beta^{\gamma_i}(z) + \sum_{j=\psi_i+1}^{\psi_i+B\gamma_k-\psi_i} \beta^{\gamma_i+j-1}(z) + \right. \\ \left. + \sum_{j=\psi_i+B\gamma_k-\psi_i+1}^k \beta^{\gamma_i+j-1}(z) \alpha^{L_{\psi_i+k}}(z) \right], & k \geq \psi_i, \end{cases} \quad (20)$$

where $\gamma_i = \left\lfloor \frac{i}{B} \right\rfloor$ - the number of completed batches formed in the system given that, at the instant a batch arrives, there are already i packets in the system,

$\gamma_{i+j-1} = \left\lfloor \frac{i+j-1}{B} \right\rfloor$ - the number of full batches formed before an arbitrary packet in the arriving batch, $j = 1, \dots, k$,

$\psi_i = B(\gamma_i + 1) - i$ - the number of available places in the batch being formed,

$L_n = \sum_{m=n}^A l_m$ - the probability that an arrival batch has at least n packets.

Finally, the stationary distribution of the sojourn time of a packet will have the following PGF:

$$V_{LA}(z) = \mathcal{W}_{LA}(z)\beta(z). \quad (21)$$

We can also calculate the average waiting time and the average sojourn time of a packet with the help of PGFs, respectively, as

$$\bar{w}_{LA} = \mathcal{W}'_{LA}(1), \quad (22)$$

$$\bar{v}_{LA} = V'_{LA}(1) = \bar{w}_{LA} + \frac{1}{b}. \quad (23)$$

As for EA system the stationary probability distribution $\{q_i^{ea}, i \in \Omega\}$ of the states of the DTMC $Q^*(n), n = 0, 1, \dots$, coincides with the stationary probability distribution $\{p_i^{ea}, i \in \Omega\}$ of the system states of the DTMC $Q^-(n), n = 0, 1, \dots$, i.e.

$$q_i^{ea} = p_i^{ea}, \quad i \geq 0. \quad (24)$$

In this case, the PGF of the waiting time of an arbitrary packet from the arrival batch and the PGF of the sojourn time of a packet in the system will be as follows, respectively,

$$\mathcal{W}_{EA}(z) = \begin{cases} \sum_{i=0}^{\infty} p_i^{ea} \sum_{k=1}^A \frac{l_k}{\hat{l}} \left[\sum_{j=1}^k \beta^{\gamma_i-1}(z) \alpha^{L_{\psi_i+k}}(z) \right], & k < \psi_i, \\ \sum_{i=0}^{\infty} p_i^{ea} \sum_{k=1}^A \frac{l_k}{\hat{l}} \left[\sum_{j=1}^{\psi_i} \beta^{\gamma_i-1}(z) + \sum_{j=\psi_i+1}^{\psi_i+B\gamma_k-\psi_i} \beta^{\gamma_{i+j-1}}(z) + \right. \\ \left. + \sum_{j=\psi_i+B\gamma_k-\psi_i+1}^k \beta^{\gamma_{i+j-1}}(z) \alpha^{L_{\psi_i+k}}(z) \right], & k \geq \psi_i, \end{cases} \quad (25)$$

$$V_{EA}(z) = \mathcal{W}_{EA}(z) \beta(z), \quad (26)$$

where ψ_i, γ_i and γ_{i+j-1} are defined in the same way as for the LA system.

In terms of average values, these take the following forms:

$$\bar{w}_{EA} = \mathcal{W}'_{EA}(1), \quad (27)$$

$$\bar{v}_{EA} = V'_{EA}(1) = \bar{w}_{EA} + \frac{1}{b}. \quad (28)$$

Additionally, for both systems we can find the average number of packets in the system:

$$N_{LA} = P'_{LA}(1) = \frac{bB(1-\rho) \left(a\hat{l} + \frac{B-1}{2} - \sum_{j=B}^{A+B} \frac{1}{1-z_j} \right)}{(-1)^{A+B} a \bar{b} l_A \prod_{j=1}^{A+B} (z_j - 1)}. \quad (29)$$

$$N_{EA} = P'_{EA}(1) = \frac{\sum_{i=0}^{B-1} p_i^{ea} \left(\bar{a} + a \sum_{k=1}^{B-i-1} l_k \right)}{1-\rho}. \quad (30)$$

That allows us to use Little's formula to deriving average system time for both cases:

$$\bar{v}_{LA} = \frac{N_{LA}}{a\hat{l}}, \quad (31)$$

$$\bar{v}_{EA} = \frac{N_{EA}}{a\hat{l}}. \quad (32)$$

2.3 Age of Information

In this section, we will define the PGF of PAoI in a general case and then we will look at specific cases.

According to [14] we define AoI at the sink as the function

$$\Delta(t) = t - t_{N(t)}, \quad (33)$$

where $N(t)$ is the indicator of the update (a packet) from the source (RD_{PT}) most recently received by the sink (RD_{FT}).

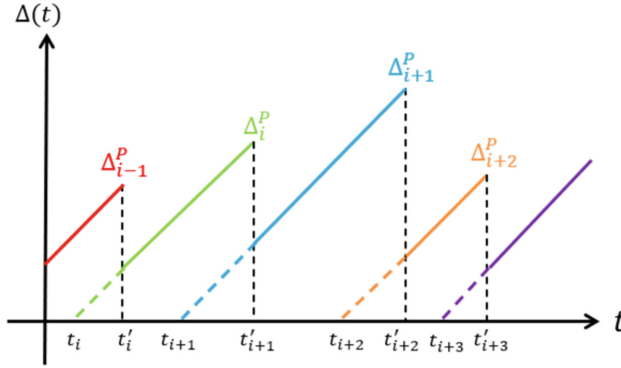


Fig. 3. Function of AoI $\Delta(t)$ and PAoI values Δ_i^P

In the Fig. 3, let $t_i, i \geq 0$, be the time instant of generating the i th packet on the source and $t'_i, i \geq 0$, be the time instant when the i th packet was received by the sink. Then we denote the peak age of information (PAoI) [15] for the i th packet at the sink as follows

$$\Delta_i^P = t'_i - t_{i-1}. \quad (34)$$

It can be noticed that PAoI is the r.v. being calculated as the sum of the r.v. ξ of the interarrival time with PGF $\alpha(z)$, the r.v. ω of the waiting time with PGF $\mathcal{W}(z)$ and the r.v. η of service time with PGF $\beta(z)$ as demonstrated in Fig. 4.

Due to that, we are able to obtain the PGF of PAoI as the PGFs convolution for the LA system

$$\Delta_{LA}^P(z) = \alpha(z)V_{LA}(z) = \alpha(z)\mathcal{W}_{LA}(z)\beta(z), \quad (35)$$

with the average PAoI

$$\bar{\Delta}_{LA}^P = \frac{1}{a\hat{l}} + \bar{v}_{LA}, \quad (36)$$

and for the EA system

$$\Delta_{EA}^P(z) = \alpha(z)V_{EA}(z) = \alpha(z)\mathcal{W}_{EA}(z)\beta(z), \quad (37)$$

with the average PAoI

$$\bar{\Delta}_{EA}^P = \frac{1}{a\hat{l}} + \bar{v}_{EA}. \quad (38)$$

3 Numerical Analysis

In this section, we estimate the impact of two-hop DECT-2020 network fragment in AoI metrics when delivering an update from RD_{PT} to RD_{FT} . For this we model a $\text{RD}_{\text{FT,PT}}$ node as LA and EA systems and compare the results of numerical experiments on both systems with parameters $A = 5$, $B = 3$, and $b = 0.8$.

We consider a node $\text{RD}_{\text{FT,PT}}$ that receives packets from 5 end-nodes RD_{PTS} , which implies $A = 5$ in the mathematical model. We assume that the batch size has a binomial distribution with a success probability of 0.7. Note that in real DECT-2020 networks the batch size distribution depends on the probability of RD_{PT} having data at each time unit and the number of RD_{PTS} in the cluster. A fixed size of the serviced batch $B = 3$ and a probability $b = 0.8$ of completing the service of a batch in a time slot allow us to regulate the load served by

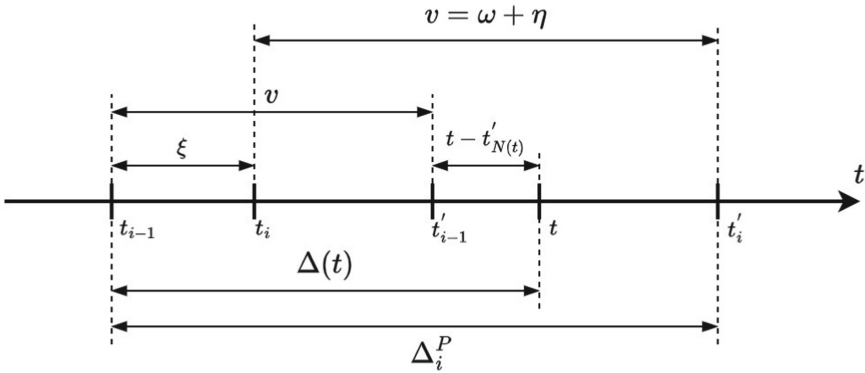


Fig. 4. Peak Age of Information Δ_i^P for the i th packet

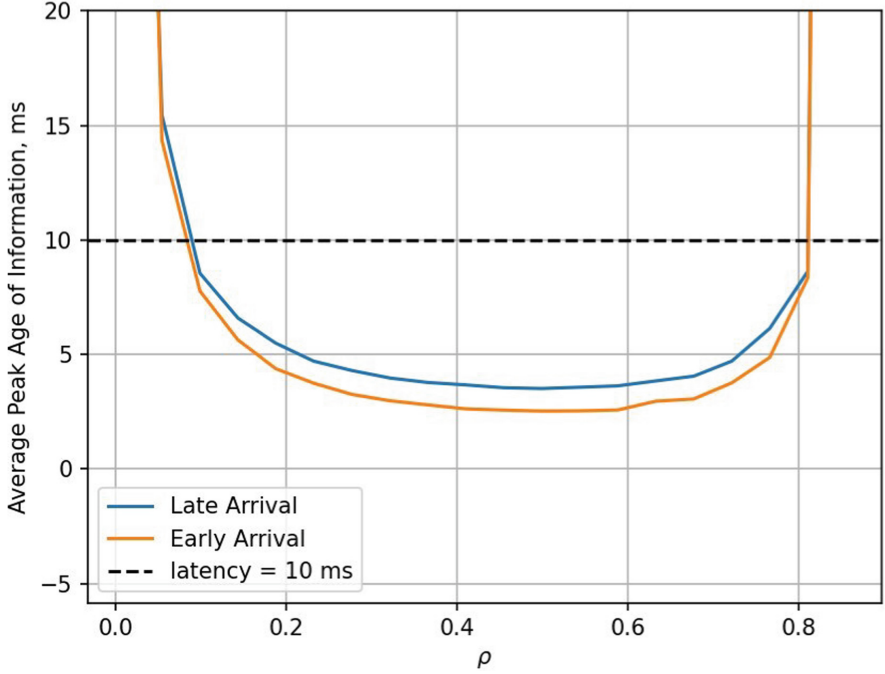


Fig. 5. The comparison EA and LA systems in case of the average PAoI

$RD_{FT,PT}$ over the time interval during which the radio channel at $RD_{FT,PT}$ is used to transmit through the uplink to the parent node the packets received from the child nodes before. We assume that in this and subsequent experiments, the packet delivery probability a is related to the values of these parameters by (1). Note that a time unit that corresponds to a slot in mathematical model of DTMC can be interpreted as a slot, subslot, or symbol in DECT-2020 protocol. In the latter case, its duration in seconds will be calculated using the Fourier transform scaling factor β and subcarrier scaling factor μ , which depend on the numerology). For our analysis we consider numerology 1 with symbol length $41.667 \cdot 10^{-6}$ s.

In Fig. 5, for the first experiment we present the average PAoI $\bar{\Delta}^P$ for LA (36) and EA (38) systems as a function of load. As a result of this experiment, we found the load range of $\rho = [0,09...0,81]$, where the system with selected range of system parameter values gives the average peak data age is less than 10 ms, that satisfies the stringent latency requirements for URLLC traffic [13].

In Fig. 6, for the second experiment we compare the average number of packets in LA (29) and EA (30) systems as a function of load using the same parameters as in the first experiment. As expected, the curve for LA lies above the curve for EA, although only slightly, so LA system performance metrics show the worse case.

The third experiment involves analyzing the average PAoI $\overline{\Delta}^P$ as a function of the ratio B/A between the length of the serviced and received batches (see Fig. 7). The experiment is carried out at $A = 10$, $B = \{1, \dots, 10\}$, $b = 0.8$ under conditions of decreasing load. As the graphs show, for $\overline{\Delta}^P$ the value of 10 ms is achieved starting from a ratio value of 0.25 and higher, and the difference for high and low $\overline{\Delta}^P$ is 0.5 - 1 ms.

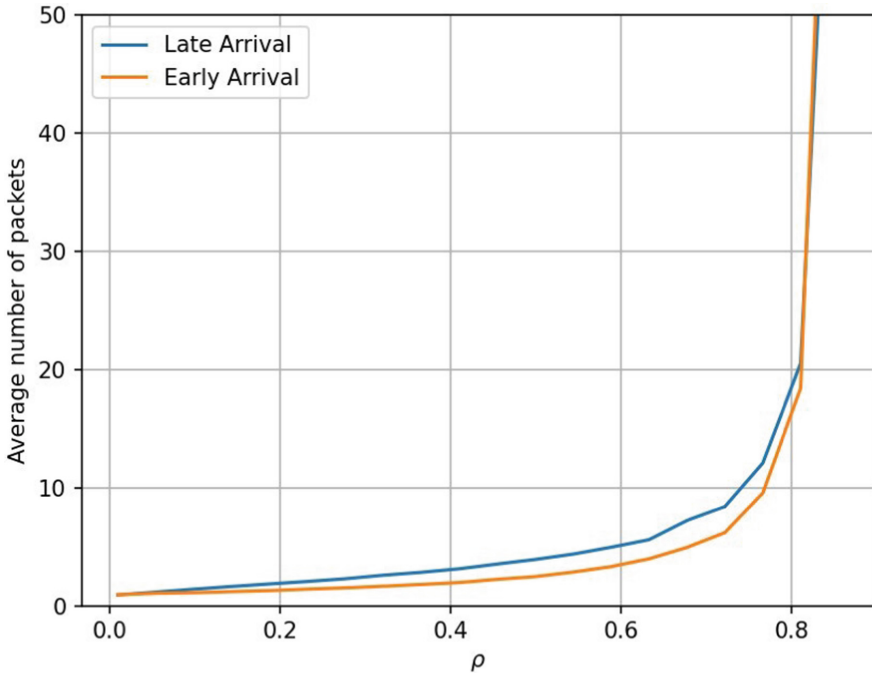


Fig. 6. Performance of average numbers of packets in the EA and LA systems

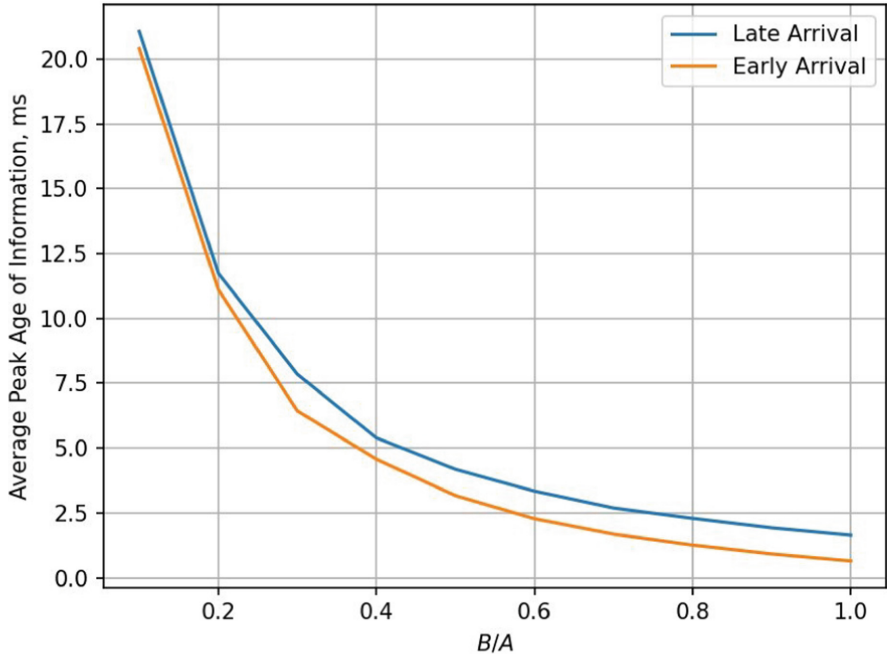


Fig. 7. Performance of dependence of the average PAoI on batches sizes

4 Conclusion

In this work, we investigated a model of a DECT-2020 NR network cluster using a discrete-time system with different arrival disciplines. Based on the derived PGF of the waiting time, we also succeed to obtain an analytical expression for the PGF of the PAoI. We compared the average peak age of information in the DECT-2020 NR network cluster under the EA and LA disciplines. As the results of experiments, it is recommended to rely on the LA system, which provides an upper bound for the delay metrics, when calculating and designing a DECT-2020 NR network. In future work, we also plan to focus on the study of discrete-time systems with batch service of variable batch size. Estimating the distribution of the incoming batch length is also the goal of our further research.

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Appendix

Derivation of the PGF of the number of packets in the system

Early Arrival System. Let us multiply (4) by z^i and sum over i , then we will have

$$\begin{aligned}
 P_{EA}(z) &= \sum_{i=0}^{B-1} z^i \left[\sum_{j=0}^{i-A+B-1} p_j^{ea} al_{i-j} + \sum_{j=\max(0, i-A+B)}^{i-1} p_j^{ea} (al_{i-j} + al_{i-j+B}b) + \right. \\
 &\quad \left. + p_i^{ea}(\bar{a} + al_Bb) + \sum_{j=i+1}^{B+i-1} p_j^{ea} al_{i-j+B}b + p_{i+B}^{ea} \bar{a}b \right] + \sum_{i=B}^{\infty} z^i \left[\sum_{j=\max(0, i-A)}^{i-A+B-1} p_j^{ea} al_{i-j} \bar{b} + \right. \\
 &\quad \left. + \sum_{j=i-A+B}^{i-1} p_j^{ea} (al_{i-j} \bar{b} + al_{i-j+B}b) + p_i^{ea}(\bar{a} \bar{b} + al_Bb) + \sum_{j=i+1}^{B+i-1} p_j^{ea} al_{i-j+B}b + p_{i+B}^{ea} \bar{a}b \right] = \\
 &= \bar{a} \bar{b} P_{EA}(z) + \sum_{i=0}^{B-1} z^i \sum_{j=0}^{i-A+B-1} p_j^{ea} al_{i-j} b + \sum_{i=0}^{B-1} z^i \sum_{j=\max(0, i-A+B)}^{i-1} p_j^{ea} al_{i-j} b + \\
 &\quad + \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{i-A+B-1} p_j^{ea} al_{i-j} \bar{b} + \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A+B)}^{i-1} p_j^{ea} al_{i-j} \bar{b} + \bar{a} b \sum_{i=0}^{B-1} z^i p_i^{ea} + \\
 &\quad + \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A+B)}^{B+i-1} p_j^{ea} al_{i-j+B}b + \bar{a} b \sum_{i=0}^{\infty} z^i p_{i+B}^{ea} = \bar{a} \bar{b} P_{EA}(z) + \bar{a} b \sum_{i=0}^{B-1} z^i p_i^{ea} + \\
 &\quad + \bar{a} b \sum_{i=0}^{B-1} z^i \sum_{j=0}^{i-1} p_j^{ea} l_{i-j} + \bar{a} b \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{i-1} p_j^{ea} l_{i-j} + \bar{a} b \sum_{i=0}^{\infty} z^i p_{i+B}^{ea} + \\
 &\quad + \bar{a} b \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A+B)}^{B+i-1} p_j^{ea} l_{i-j+B}.
 \end{aligned}$$

Let us look at each expression separately:

- 1) $ab \sum_{i=0}^{B-1} z^i \sum_{j=0}^{i-1} p_j^{ea} l_{i-j} = ab \sum_{i=0}^{B-2} z^i p_i^{ea} \sum_{k=1}^{B-i-1} z^k l_k,$
- 2) $\bar{a} b \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{i-1} p_j^{ea} l_{i-j} = \bar{a} b L(z) P_{EA}(z),$
- 3) $ab \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A+B)}^{B+i-1} p_j^{ea} l_{i-j+B} = \frac{ab}{z^B} \left(L(z) P_{EA}(z) - \sum_{i=0}^{B-1} z^i p_i^{ea} \sum_{k=1}^{B-i-1} z^k l_k \right),$
- 4) $\bar{a} b \sum_{i=0}^{\infty} z^i p_{i+B}^{ea} = \frac{\bar{a} b}{z^B} \left(L(z) P_{EA}(z) - \sum_{i=0}^{B-1} z^i p_i^{ea} \right).$

Then, the final form of the PGF is:

$$P_{EA}(z) = \frac{(\bar{a} b - \frac{\bar{a} b}{z^B}) \sum_{i=0}^{B-1} z^i p_i^{ea} + (ab - \frac{ab}{z^B}) \sum_{i=0}^{B-1} z^i p_i^{ea} \sum_{k=1}^{B-i-1} z^k l_k}{1 - (\bar{a} \bar{b} + \bar{a} b L(z) + \frac{ab}{z^B} L(z) + \frac{\bar{a} b}{z^B} L(z))},$$

which is equivalent (6).

Late Arrival System. After multiplying (10) by z^i and summing up over i , we have

$$\begin{aligned}
P_{LA}(z) &= \sum_{i=0}^{B-1} z^i \left[\sum_{j=0}^{i-1} p_j^{la} a_{i-j} + p_i^{la} \bar{a} + \sum_{j=B}^{B+i-1} p_j^{la} b a_{i-j+B} + p_{i+B}^{la} b \bar{a} \right] + \\
&+ \sum_{i=B}^A z^i \left[\sum_{j=0}^{B-1} p_j^{la} a_{i-j} + \sum_{j=B}^{i-1} p_j^{la} (\bar{b} a_{i-j} + b a_{i-j+B}) + p_i^{la} (\bar{b} \bar{a} + b a_B) + \right. \\
&+ \left. \sum_{j=i+1}^{B+i-1} p_j^{la} b a_{i-j+B} + p_{i+B}^{la} b \bar{a} \right] + \sum_{i=0}^{\infty} z^i \left[\sum_{j=i-A}^{B-1} p_j^{la} a_{i-j} + \sum_{j=\max(B, i-A)}^{i-A+B-1} p_j^{la} \bar{b} a_{i-j} + \right. \\
&+ \left. \sum_{j=i-A+B}^{i-1} p_j^{la} (\bar{b} a_{i-j} + b a_{i-j+B}) + p_i^{la} (\bar{b} \bar{a} + b a_B) + \sum_{j=i+1}^{B+i-1} p_j^{la} b a_{i-j+B} + p_{i+B}^{la} b \bar{a} \right] = \\
&= \bar{b} \bar{a} \sum_{i=0}^{\infty} z^i p_i^{la} + \bar{a} b \sum_{i=0}^{B-1} z^i p_i^{la} + \bar{a} b \sum_{i=0}^{\infty} z^i p_{B+i}^{la} + \sum_{i=0}^A z^i \sum_{j=B}^{B+i-1} p_j^{la} b a_{i-j+B} + \\
&+ \sum_{i=A+1}^{\infty} z^i \sum_{j=i-A+B}^{B+i-1} p_j^{la} b a_{i-j+B} + \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{\min(i-1, B-1)} p_j^{la} a_{i-j} + \sum_{i=B}^A z^i \sum_{j=B}^{i-1} p_j^{la} \bar{b} a_{i-j} + \\
&+ \sum_{i=A+1}^{\infty} z^i \sum_{j=\max(B, i-A)}^{i-1} p_j^{la} \bar{b} a_{i-j} = \bar{a} \bar{b} P_{LA}(z) + \bar{a} b \sum_{i=0}^{B-1} z^i p_i^{la} + \bar{a} b \sum_{i=0}^{\infty} z^i p_{B+i}^{la} + \\
&+ \sum_{i=0}^{\infty} z^i \sum_{j=\max(B, i-A+B)}^{B+i-1} p_j^{la} b a_{i-j+B} + \sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{\min(i-1, B-1)} p_j^{la} a_{i-j} + \\
&+ \sum_{i=0}^{\infty} z^i \sum_{j=\max(B, i-A)}^{i-1} p_j^{la} \bar{b} a_{i-j}.
\end{aligned}$$

Let us analyze these formulas separately.

- 1) $\bar{a} b \sum_{i=0}^{\infty} z^i p_{B+i}^{la} = \frac{\bar{a} b}{z^B} \left(P_{LA}(z) - \sum_{i=0}^{B-1} z^i p_i^{la} \right),$
- 2) $\sum_{i=0}^{\infty} z^i \sum_{j=\max(B, i-A+B)}^{B+i-1} p_j^{la} b a_{i-j+B} = b a L(z) \sum_{i=0}^{\infty} z^i p_{i+B}^{la} = \frac{b a L(z)}{z^B} \left(P_{LA}(z) - \sum_{i=0}^{B-1} z^i p_i^{la} \right),$
- 3) $\sum_{i=0}^{\infty} z^i \sum_{j=\max(0, i-A)}^{\min(i-1, B-1)} p_j^{la} a_{i-j} = \sum_{i=0}^{A+B-1} z^i \sum_{j=\max(0, i-A)}^{\min(i-1, B-1)} p_j^{la} a_{i-j} = a L(z) \sum_{i=0}^{B-1} z^i p_i^{la},$
- 4) $\sum_{i=0}^{\infty} z^i \sum_{j=\max(B, i-A)}^{i-1} p_j^{la} \bar{b} a_{i-j} = \sum_{i=B}^{\infty} z^i \sum_{j=\max(B, i-A)}^{i-1} p_j^{la} \bar{b} a_{i-j} = \bar{b} a L(z) \left(P_{LA}(z) - \sum_{i=0}^{B-1} z^i p_i^{la} \right).$

Then the PGF will have the following form

$$P_{LA}(z) = \frac{\sum_{i=0}^{B-1} z^i p_i^{la} \left(\bar{a}b - \frac{\bar{a}b}{z^B} - \frac{baL(z)}{z^B} + aL(z) - \bar{b}aL(z) \right)}{1 - \left(\bar{a}b + \frac{\bar{a}b}{z^B} + \frac{baL(z)}{z^B} + \bar{b}aL(z) \right)}.$$

After simple mathematical operations, we finally obtained (12).

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